

Course milestone M2: Your Rules and Your Question

HONR 46400 · Evidence-Driven Research

What to submit

One file, `lastname_m02.pdf`. Everything this milestone asks for goes inside it, as sections in the order below.

Submit it on Brightspace, under **Assignments** then **M02**. Brightspace carries the deadline.

Your work lives in a Colab notebook, which has no export-to-PDF button. **Making the PDF you hand in**, at the end of this document, is the one-minute routine that turns the notebook into the file you upload.

This milestone presents **Book Milestone 2: Your rules and your question** (checkpoint, version 1) of the course book, EDR|AI.

Open the milestone workbook in Colab

This chapter closes Studio 2: Set your rules, shape your question. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. Your working agreement (an ownership statement, your working rules, your opened AI Research Ledger, your delegation map, and the red-flag screen of your chosen problem) and your question, formally declared: objective, units, outcomes and conditions, kind and reach, and a provisional claim boundary.

What you bring

You also carry forward the last milestone's artifact: **Milestone 1: Your curiosity, committed**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 2: AI Is Your Arm, Not Your Brain: your opened AI Research Ledger and one verified factual hinge from your own problem, including the retrospective Studio 1 receipt as row one.
- Lesson 3: You as a Research Director: your delegation map for the whole project and for the declaration ahead: candidate wording may travel, while the lead question, kind, reach, and boundary stay yours.
- Lesson 4: Specify, Delegate, Interrogate, Inspect, Verify, Document, Defend: one complete SDIIVDD run on a checkable claim your starting belief makes, with the check's outcome on record: including "nothing changed" when the check agreed.
- Lesson 5: Research Responsibility and Intellectual Ownership: your ownership statement, never-delegate list, and first AI-use disclosure, with the research question named as the first sentence your name signs.
- Lesson 6: Choose Your Question's Kind and Reach: your lead question chosen from the candidates, placed by kind and reach, with double-barrels split and a comparison world recorded when causal.
- Lesson 7: Declare Your Research Question: your chosen formal wording with every change accounted for, the field card, the boundary pair with its uncertainty-and-limitations line, and the stranger-test result, dated as version zero.

The practice

1. Write your ownership statement: the decisions that stay yours no matter how good the tool gets, the choice of question first among them.
2. Write your working rules: what you will delegate, what you will always verify, and how you will record both. Mark the declaration work on the map: candidate wordings are delegable; kind, reach, and the boundary never are.
3. Open your AI Research Ledger and give its first rows to the exchanges you have already run, the Studio 1 brainstorm included.
4. Build and run the red-flag screen on your chosen problem: is it answerable, is it yours to ask, and could you defend attempting it.

5. Assemble your declaration from the last two lessons: the lead question classified by kind and reach, the field card (objective, unit, outcome, conditions), and the chosen wording with every change accounted for.
6. Write your provisional claim boundary beside it: the sentence you hope to defend, the stronger sentence you will not be able to, and the uncertainty-and-limitations line any answer will have to carry.
7. Read the two artifacts against each other: every never-delegate line in the agreement must cover a decision the declaration actually made. Where one names a rule the other ignored, fix the weaker document.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** The red-flag screen runs here, and a problem that studies people leaves this studio with that fact attached.
- **Evidence, provenance, and reproducibility.** Nothing here is settled until Studio 3 checks it against what is already known.
- **AI activity, verification, and human decisions.** The ledger opens here and every earlier exchange gets its row; from now on, no delegation goes unrecorded.
- **Uncertainty, claim boundary, and revision history.** A question with no recognizable answer cannot carry uncertainty; fix the question before you leave.

Where this milestone sits

You arrived carrying Milestone 1: Your curiosity, committed. Milestone 3 holds the declared question up against what is already known, and it will usually send back a sharper version. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

Studio 3 almost always sends the question back here for a sharper version, and any change in your tools, your data source, or your collaborators reopens the agreement.

The book's own checks

Your course rubric on Brightspace scores these checks inside its rows, so use this table to test your work before you hand it in. The points below are the book's, not your course score.

Each row scores **0, 1, or 2. 12 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning

#	Criterion	0	1	2
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: Your working agreement (responsibility statement, working rules, opened AI Research Ledger, delegation map, and the red-flag screen of your chosen problem) and your question, formally declared: objective, units, outcomes and conditions, kind and reach, and a provisional claim boundary	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated
6	The declared question is fixed by its own words: kind and reach match the sentence, double-barrels are split, and the claim boundary states the defensible sentence, the forbidden sentence, and the uncertainty-and-limitations line the answer must carry	no declaration, or a kind or reach the question's words do not fix	declared, but one term is loose enough that a stranger could not say what gets recorded or who the answer is about	declared so a stranger could restate unit, record, and audience, with both boundary sentences present

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.

Making the PDF you hand in

Colab has no export-to-PDF button. You make the PDF by **printing the notebook to a file**, which takes about a minute once you know the preparation steps. Skipping them is the most common way a milestone arrives looking half finished, because a notebook prints what is on the screen and nothing else.

1. Prepare the notebook

1. **Run the whole thing.** `Runtime` → `Run all`, then wait for it to finish. A cell you never ran prints with nothing underneath it, so an analysis you really did can reach the grader as an empty box.
2. **Expand every collapsed section.** Anything folded shut prints as a heading with nothing beneath it. Open all of them, including any Colab folded for you.
3. **Shorten any output that scrolls.** When Colab puts a scrollbar on an output, or tells you the output is truncated, only the visible part prints. Show the first twenty rows instead of the whole table, or summarise it, then rerun that cell. A reader did not want the whole table anyway.
4. **Read it once, top to bottom.** You are about to freeze it.

2. Print it to a file

`File` → `Print` opens your browser's print dialog. Set the destination to **Save as PDF**, open the dialog's further settings, and turn on **Background graphics**, so code cells keep their shading and your figures keep their backgrounds. Save it under the file name this milestone asks for.

Wide tables and wide figures are the usual casualty. If something runs off the right edge, switch the layout to **landscape** or reduce the scale, and print again.

3. Check the PDF before you upload it

Open the file you just made and confirm three things: every figure is whole rather than sliced by a page break, no output stops mid-line, and the pages run in order with nothing missing. A PDF is an artifact like any other in this course, so you verify it before your name goes on it.

One route not to take

Ask an AI assistant how to export a Colab notebook and it will very likely hand you `jupyter nbconvert --to pdf`. Do not spend your afternoon on it. Inside Colab that route needs a large LaTeX installation, runs for several minutes, and then either fails or quietly drops the symbols and emoji these notebooks are full of. Printing to a file is the route that works.

That is a small, cheap instance of the rule this whole course runs on. The confident answer is not automatically the correct one, and the way you find out is to check it against what actually happens.